

Main Structure: First by fibration theory, we show that $\pi_i U(n)$ is stable, and that $\pi_i U(n) \cong \pi_i SU(n)$. Also by fibrations we connect $\pi_i U(2m)$ with $\pi_i G_m(\mathbb{C}^{2m})$. Finally, using Morse theory and by studying the structure of $U(n)$, we find the connection of $SU(n)$ and $G_m(\mathbb{C}^{2m})$.

Let $U(n)$ be the set of unitary matrices in $\mathbb{C}^{n \times n}$ with standard inner product, then $U(n)$ is a compact Lie group.

Lemma: The tangent space $TU(n)_I$ can be identified with the space of skew-Hermitian matrices.

Proof: Consider $\exp : \mathbb{C}^{n \times n} \rightarrow \mathbb{C}^{n \times n}, A \mapsto \sum_{n=0}^{\infty} \frac{1}{n!} A^n$, which is a local homeomorphism from $0 \in \mathfrak{u}(n) = \{A \in \mathbb{C}^{n \times n} : A + A^* = 0\}$ to $I \in U(n)$.

If $A + A^* = 0$, then take $\gamma : [0, 1] \rightarrow U(n), t \mapsto \exp(tA)$, we have $\exp(tA) \in U(n)$ since $(\exp(tA))^* = \exp(tA^*) = \exp(-tA)$. Also $\gamma(0) = I$ and $\gamma'(0) = A$, so $A \in TU(n)_I$.

For any $A \in TU(n)_I$, there exists $\gamma : [0, 1] \rightarrow U(n)$ such that $\gamma(0) = I$ and $\gamma'(0) = A$. Differentiating $\gamma(t)^* \gamma(t) = I$, we have $\gamma(t)' \gamma(t) + \gamma(t)^* \gamma'(t) = 0$, and at $t = 0, A^* + A = 0$. $\square$

Hence the Lie algebra $\mathfrak{g}$ of $U(n)$ can also be identified with skew-Hermitian matrices, since for any $A, B \in TU(n)_I, [A, B] = AB - BA$ corresponds to the bracket product of $\exp(tA)$ and $\exp(tB)$.

Note

Construction of Riemannian Metric:

We explicitly construct a Riemannian metric on $U(n)$:

For matrices $A, B \in \mathfrak{g}$, define $\langle A, B \rangle = \operatorname{Re} \operatorname{tr}(AB^*)$, then it is an inner product, so it defines a unique left invariant Riemannian metric. To verify the metric is also right invariant, we check that it is invariant under $A \mapsto SAS^{-1}$.

The induced linear map is $\operatorname{Ad}(S) : TU(n)_I \rightarrow TU(n)_I$, given by $\operatorname{Ad}(S)A = SAS^{-1}$ since $\exp(SAS^{-1}) = S \exp(A) S^{-1}$. Hence the inner product is clearly invariant under $\operatorname{Ad}(S)$.

Observe that any $A \in \mathfrak{g}$ can be unitarily diagonalized as $TAT^{-1} = \operatorname{diag}(ia_1, \dots, ia_n)$, and any $S \in U(n)$ can be unitarily diagonalized as $TST^{-1} = \operatorname{diag}(e^{ia_1}, \dots, e^{ia_n})$. Hence $\exp : \mathfrak{g} \rightarrow U(n)$ is surjective.

Analogously, for $\mathfrak{g}' = \{A : A + A^* = 0, \operatorname{tr}(A) = 0\}$, the identity $e^{\operatorname{tr}(A)} = \det(\exp A)$ for $A \in \mathfrak{g}$ implies $\mathfrak{g}'$ is the Lie algebra for $SU(n)$, since $\det(\exp tA) = 1$ for all small t implies $\operatorname{tr}(A) = 0$.

Now we consider all geodesics of $U(n)$ from I to $-I$, i.e. all $A \in \mathfrak{g}$ such that $\exp(A) = -I$. Take $T \in U(n)$ such that TAT^{-1} is diagonal, then $\exp(TAT^{-1}) = T(-I)T^{-1} = -I$, so $\exp(A) = -I$ iff the eigenvalues of A are in the form $\{k_1 i\pi, \dots, k_n i\pi\}$ where k_i are odd integers.

The length of the geodesic $t \mapsto \exp(tA)$ is $|A| = \sqrt{\operatorname{tr}(AA^*)} = \pi \sqrt{\sum k_i^2}$, so A is a minimal geodesic iff $k_i \in \{\pm 1\}$. In this case, A is uniquely determined by $\operatorname{Ker}(A - i\pi I)$, which is an arbitrary subspace of $\mathbb{C}^n$.

Hence the minimal geodesics from I to $-I$ correspond to subspaces of $\mathbb{C}^n$.

To make it simple, consider instead $SU(n)$ where $n = 2m$, then with the requirement $\operatorname{tr}(A) = 0$, $\operatorname{Ker}(A - i\pi I)$ must be an m -dimensional subspace of $\mathbb{C}^{2m}$, so the space of geodesics is homeomorphic to the Grassmann manifold $G_m(\mathbb{C}^{2m})$.

Tip

Grassmann Manifold:

The Grassmann manifold $G_m(\mathbb{C}^{2m})$ consists of (W, V) such that $\mathbb{C}^{2m} = W \oplus V$ is an orthogonal direct sum, with the relative topology.

For each $(W_0, V_0) \in G_m(\mathbb{C}^{2m})$, consider all m -subspaces W such that $W \cap V_0 = 0$, then they correspond to linear maps $T : W_0 \rightarrow V_0$ where $W = \{w + T(w) : w \in W_0\}$. Hence this open set is homeomorphic to $\text{Hom}_{\mathbb{C}}(W_0, V_0) \cong \mathbb{C}^{m^2}$, so we obtain a coordinate chart.

The transition charts are smooth (indeed holomorphic), so $G_m(\mathbb{C}^{2m})$ is a complex smooth manifold.

We will prove the following lemma:

Lemma: Any non-minimal geodesic in $\Omega(SU(2m); I, -I)$ has index at least $2m + 2$.

Since the $(n + 1)$ -skeleton X^{n+1} completely determines $\pi_n(X)$, by Morse Index Theorem we obtain:

Theorem: $\pi_i(G_m(\mathbb{C}^{2m})) \cong \pi_i(\Omega(SU(2m))) \cong \pi_{i+1}(SU(2m))$ for all $i \leq 2m$.

Note

Fibrations and long exact sequences:

Def: A **fibration** $p : E \rightarrow B$ is a map that satisfy the homotopy lifting property: for any homotopy $h : X \times [0, 1] \rightarrow B$ and $\tilde{h}_0 : X \times \{0\} \rightarrow E$ lifting $h|_{X \times \{0\}}$, there exists (not necessarily unique) a homotopy $\tilde{h} : X \times [0, 1] \rightarrow E$ lifting h such that $\tilde{h}|_{X \times \{0\}} = \tilde{h}_0$.

Given a fibration $p : E \rightarrow B$ and any $b_0 \in B$, the fiber is $F = p^{-1}(b_0)$.

Analogous to exact sequences of chain complexes, a fibration induces a long exact sequence of homotopy groups.

Theorem: A fibration $F \hookrightarrow E \xrightarrow{p} B$ induces a long exact sequence

$$\cdots \rightarrow \pi_i(F) \xrightarrow{\iota_*} \pi_i(E) \xrightarrow{p_*} \pi_i(B) \xrightarrow{\partial} \pi_{i-1}(F) \rightarrow \cdots$$

Proof: By homotopy lifting property, any fibration induces the exact sequence $\pi_i(F) \rightarrow \pi_i(E) \rightarrow \pi_i(B)$. Also by this property, there is a natural map from the path space $\Omega B \rightarrow F$ such that $\Omega B \rightarrow F \rightarrow E$ is homotopy equivalent to a fibration. Hence we obtain the Puppe sequence

$$\cdots \rightarrow \Omega^2 B \rightarrow \Omega F \rightarrow \Omega E \rightarrow \Omega B \rightarrow F \rightarrow E \rightarrow B$$

and by the natural isomorphism $\pi_n(X) \cong \pi_{n-1}(\Omega X)$, the required long exact sequence.

Using the theory above, we obtain the following isomorphisms:

For each m consider the fibration $p : U(m + 1) \rightarrow \mathbb{S}^{2m+1}, (C_1, \dots, C_{m+1}) \mapsto C_{m+1}$ with fiber $p^{-1}(e_{2m+1}) = U(m)$. It induces the long exact sequence

$$\cdots \rightarrow \pi_i U(m) \rightarrow \pi_i U_{m+1} \rightarrow \pi_i \mathbb{S}^{2m+1} \rightarrow \pi_{i-1} U(m) \rightarrow \cdots$$

since $\pi_i \mathbb{S}^{2m+1} = 0 \forall i \leq 2m$, for any $i \leq 2m - 1$ we have $\pi_i U(m) \cong \pi_i U(m + 1) \cong \pi_i U(m + 2) \cong \cdots$ so we denote the stable homotopy group $\pi_i U = \pi_i U(m)$. Furthermore, $\pi_{2m} U(m) \rightarrow \pi_{2m} U(m + 1)$ is surjective.

The fibration $p : U(m) \rightarrow \mathbb{S}^1, A \mapsto \det A$ with fiber $p^{-1}(1) = SU(m)$ gives the isomorphism $\pi_i U(m) \cong \pi_i SU(m)$ for $i \geq 2$.

By $\pi_i U(m) \cong \pi_i U(2m)$, the fibration $U(m) \rightarrow U(2m) \rightarrow U(2m)/U(m)$ (identify $U(m)$ as a subspace $A \mapsto \text{diag}(A, I)$) gives the exact sequence $\pi_i U(m) \rightarrow \pi_i U(2m) \rightarrow \pi_i(U(2m)/U(m))$ hence $\pi_i(U(2m)/U(m)) = 0$.

Note that the Grassmann manifold $G_m(\mathbb{C}^{2m})$ can be identified as $U(2m)/(U(m) \times U(m))$, so the fibration $U(m) \rightarrow U(2m)/U(m) \rightarrow U(2m)/(U(m) \times U(m))$ gives isomorphisms $\pi_{i-1} U(m) \cong \pi_i G_m(\mathbb{C}^{2m})$. Therefore,

$$\pi_{i-1} U = \pi_{i-1} U(m) \cong \pi_i G_m(\mathbb{C}^{2m}) \cong \pi_{i+1} SU(2m) \cong \pi_{i+1} U(2m) = \pi_{i+1} U.$$

Since $U(1) = \mathbb{S}^1$, $\pi_0 U = \pi_0 \mathbb{S}^1 = 0$ and $\pi_1 U = \pi_1 \mathbb{S}^1 \cong \mathbb{Z}$, therefore

Theorem (Bott): The stable homotopy groups $\pi_i U$ of $U(n)$ are periodic with period 2, and

$$\pi_{2n} U = 0, \quad \pi_{2n+1} U = \mathbb{Z}.$$

Proof of main lemma:

Lemma: Any non-minimal geodesic in $\Omega(SU(2m); I, -I)$ has index at least $2m + 2$.

Proof: Recall the Lie algebra of $SU(n)$ ($n = 2m$) is $\mathfrak{g}'$ consisting of all skew-Hermitian matrices of trace zero, and a given $A \in \mathfrak{g}'$ corresponds to a minimal geodesic from I to $-I$ iff the eigenvalues are $i\pi k_1, \dots, i\pi k_n$ where k_j are odd integers with sum zero.

Recall the conjugate points to I on the geodesic $t \mapsto \exp(tA)$ is given by the negative eigenvalues of $K_A : \mathfrak{g}' \rightarrow \mathfrak{g}', W \mapsto \mathcal{R}(A, W)A$.

Assume $A = \text{diag}(i\pi k_1, \dots, i\pi k_n)$ with $k_1 \geq \dots \geq k_n$, then if $W = (w_{ij})$, we have

$$K_A(W) = -\frac{1}{4}[[A, W], A] = -\frac{\pi^2}{4}((k_i - k_j)^2 w_{ij})$$

Note that: for each $j < l$, the matrix E_{jl} with i at (j, l) and $-i$ at (l, j) , is in $\mathfrak{g}'$ and an eigenvector of eigenvalue $-\frac{\pi^2}{4}(k_j - k_l)^2$; each diagonal matrix of trace zero is an eigenvector of eigenvalue 0.

Hence the negative eigenvalues of K_A are $-\frac{\pi^2}{4}(k_j - k_l)^2$ with $k_j > k_l$, and the conjugate points of the geodesic $t \mapsto \exp(tA)$ are exactly

$$t = \frac{2}{k_j - k_l}, \frac{4}{k_j - k_l}, \frac{6}{k_j - k_l}, \dots$$

which contributes $2 \left(\frac{k_j - k_l}{2} - 1 \right)$ to the index. Therefore

$$\lambda = \sum_{k_j > k_l} k_j - k_l - 2.$$

If γ is minimal then $k_j \in \{\pm 1\}$ so $\lambda = 0$ as expected. Otherwise for $n = 2m$

Case1: At least $m + 1$ of k_j are negative, then at least one $k_j \geq 3$ so $\lambda \geq (m + 1)(3 + 1 - 2) = 2m + 2$.

Case2: There are m positive and m negative k_j , but not all are ± 1 . Then at least one $k_j \geq 3$ and $k_l \leq -3$, so $\lambda \geq 2(m - 1) + 2(m - 1) + (3 + 3 - 2) = 2m + 2$.

Therefore in both cases the index $\lambda \geq 2m + 2$. $\square$

A note on Real Bott Periodicity:

The proof of stability and the Grassmannian carries over to $O(n)$ smoothly:

By the fibration $O(n - 1) \rightarrow O(n) \rightarrow \mathbb{S}^n$ we have $\pi_i O(n - 1) \cong \pi_i O(n)$.

By the fibration $SO(n) \rightarrow O(n) \rightarrow \mathbb{S}$ we have $\pi_i SO(n) \cong \pi_i O(n)$.

By $O(m) \rightarrow O(2m) \rightarrow O(2m)/O(m)$ we have $\pi_i(O(2m)/O(m)) = 0$.

By $O(m) \rightarrow O(2m)/O(m) \rightarrow G_m(\mathbb{R}^{2m})$ we have $\pi_i G_m(\mathbb{R}^{2m}) \cong \pi_{i-1} O(m)$.

Main failure: $\pi_i G_m(\mathbb{C}^{2m}) \cong \pi_{i+1} SU(2m)$ does not hold in the real case.

Here, the space of minimal geodesics is not $G_m(\mathbb{R}^{2m})$ but $SO(2m)/U(m)$.

Instead, Morse theory gives:

$$\begin{aligned} \Omega SO(2n) &\simeq SO(2n)/U(n) & \Omega(SO(2n)/U(n)) &\simeq U(n)/Sp(n/2) \\ \Omega(U(2n)/Sp(n)) &\simeq BSp(n) & \Omega(BSp(n)) &\simeq Sp(n) \\ \Omega Sp(n) &\simeq Sp(n)/U(n) & \Omega(Sp(n)/U(n)) &\simeq U(n)/O(n) \\ \Omega(U(n)/O(n)) &\simeq BO(n) & \Omega BO(n) &\simeq O(n). \end{aligned}$$

Therefore

$$\begin{aligned} \pi_i O &= \pi_{i+1} BO = \pi_{i+2} U/O = \pi_{i+3} Sp/U = \pi_{i+4} Sp \\ &= \pi_{i+5} BSp = \pi_{i+6} U/Sp = \pi_{i+7} SO/U = \pi_{i+8} SO = \pi_{i+8} O. \end{aligned}$$